HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented liquid and lyophilized vaccine formulations for enhanced thermal stability, successfully reducing sample degradation by **~40%** under accelerated storage conditions.
- Developed and verified thermostabilising processes for a live virus vector vaccine, ensuring critical quality attributes were maintained through **multiple freeze-thaw cycles**.
- Conducted detailed laboratory analysis, including HPLC and ELISA assays, to evaluate the efficacy of novel excipients in preserving vaccine potency and integrity.
- Managed and maintained laboratory safety protocols and data integrity practices, ensuring compliance with **Good Laboratory Practice (GLP)** standards.
- Collaborated with external manufacturing partners to transfer formulation development data and processes, facilitating scale-up activities for **preclinical and clinical trials**.
- Mentored and provided technical guidance to junior research assistants and early-career scientists on formulation development techniques and analytical methodologies.
- Maintained electronic laboratory notebooks with meticulous record-keeping for all experimental procedures, results, and observations, supporting **audit trail requirements**.

SKILLS

Vaccine Development – Formulation, Stabilization, Live Virus Vector, Lassa Fever Vaccine, Preclinical Testing, Clinical Evaluation, Biopharmaceutical

Analytical Techniques – HPLC, ELISA, Freeze-Thaw, Thermal Stability, Lyophilization, Liquid Formulation

Research & Development – Process Development, Scale-Up, Data Integrity, GLP, Research Integrity, Health and Safety

Project Management – Research Planning, Execution, Reporting, Collaboration, Mentoring, Administrative Work

Data Analysis & Reporting – Statistical Analysis, Report Preparation, Presentation, Electronic Notebooks, Data Integrity

PROJECTS

Thermostable Lassa Vaccine Formulation (NIBRT Research)

- Investigated novel cryoprotectants and lyoprotectants to develop a thermostable liquid formulation for a live Lassa virus vector vaccine, achieving **a 3-month shelf-life at 2-8°C**.
- Optimized lyophilization cycles by adjusting chamber pressure and shelf temperature profiles, resulting in improved cake structure and reconstitution times for the dried vaccine.
- Performed stability studies under various temperature and humidity conditions, correlating formulation composition with antigenicity and infectivity loss.

Vaccine Process Optimization for Scalability (NIBRT Research)

- Developed a scalable drying process for a recombinant virus vector vaccine, focusing on parameters amenable to industrial manufacturing equipment.
- Collaborated with NIBRT's biopharmaceutical manufacturing partners to validate process parameters and ensure successful technology transfer.
- Authored detailed technical reports and presentations for consortium partners and the funding body on process development findings.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)